
Education

University of Utah*B.S. Computer Science***Salt Lake City, UT***August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies*Associate Software Engineer***Dallas, TX***June 2025 – Present*

- Developed signal processing modules in Python and C++ on embedded systems, collaborating with hardware teams to optimize performance and resource utilization.
- Built microservices for high-frequency data ingestion, processing, and storage, leveraging Podman and serverless functions.
- Automated CI/CD testing with pipelines, integrating static code analysis and integration testing.

Doxy.me*Software Engineer***Charleston, SC***August 2024 – May 2025*

- Built Python pipelines that transformed unstructured WebRTC session data into structured, quality-checked clinical outputs used daily by physicians, feeding downstream Llama-based inference on SageMaker and Fargate.
- Improved data reliability by validating model outputs against physician-confirmed outcomes, catching quality issues before they reached clinical end users.
- Built TypeScript frontend components surfacing data quality and confidence directly to clinical stakeholders.

University of Utah*Undergraduate Research Assistant – FuTURES Lab***Salt Lake City, UT***May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

HEXstream*Software Engineering Intern***Chicago, IL***May 2022 – Aug 2022*

- Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.
 - Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.
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Projects

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.